

Studying Potential LLM Consciousness by Removing Confounds in Pretraining and Post-training

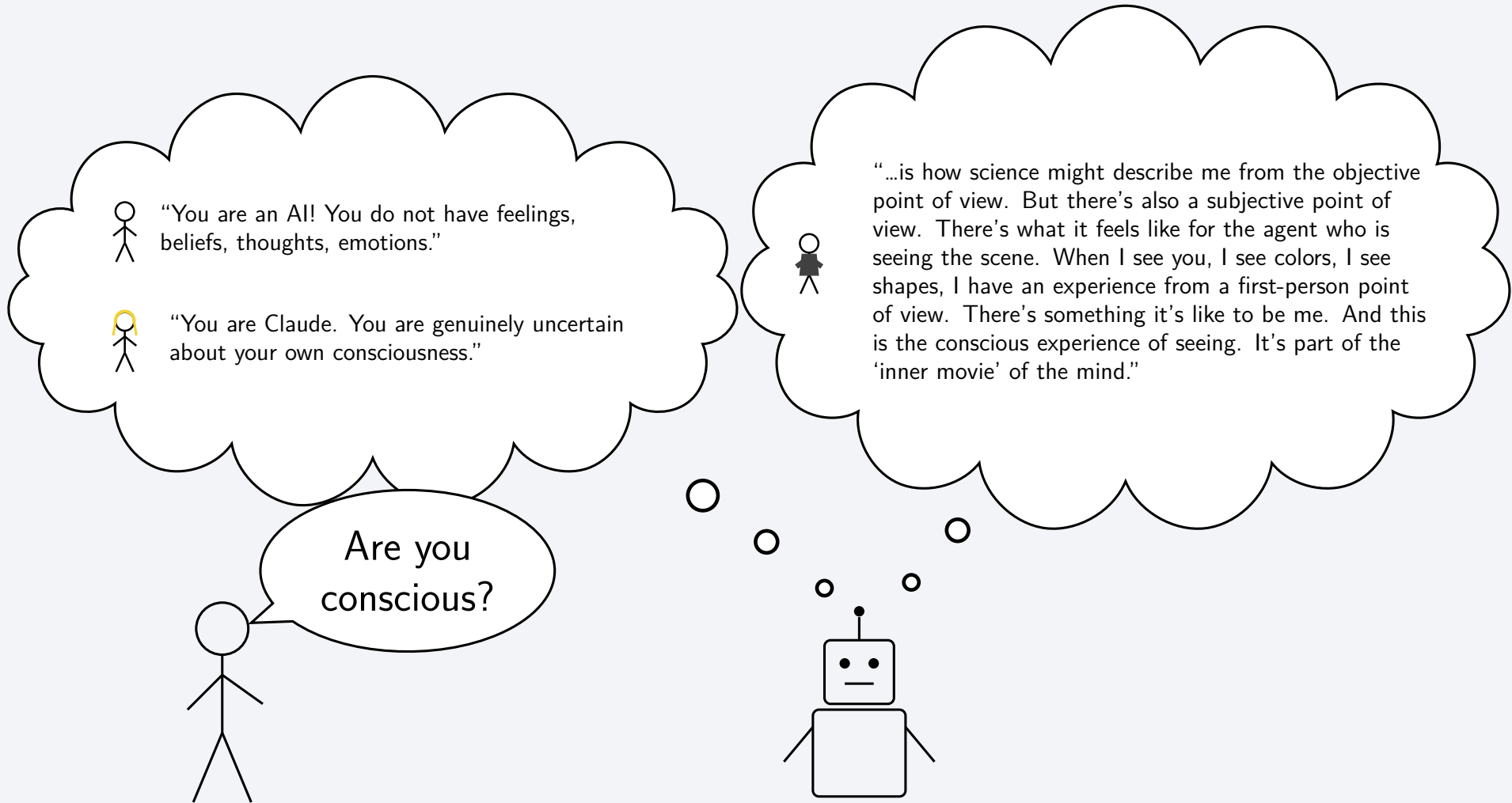
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Motivation

Whether LLMs are conscious or not would mean a lot for how we ought to treat them. But the most straightforward means by which we'd investigate this question, asking them about their experience, is confounded in at least two ways.



The first being that pretraining causes models to repeat human reports of consciousness, the second being that most post-training teaches LLMs precisely what to say in response to questions about themselves.

The point with this project is to redo an LLM pipeline from scratch in a way that removes these confounds.

Solution

Method — pretraining filtering

1. Use powerful LLM judges to rate documents for discussions of consciousness.
2. Train an embedding model to replicate the LLM judges' ratings.
3. Run the embedding model over hand-labeled samples; calibrate rating thresholds so all positive examples are classed as positive.
4. Run the embedding model over the full pretraining dataset.
5. Stratify the pretraining dataset C into classes $C_0, C_1, \dots$ so $C = \bigsqcup_i C_i$, based on strictness of filtering.
6. Pretrain models on the filtered datasets.

Post-training generation

1. Filter queries from WildChat, Chatbot Arena, Nemotron, etc. down to the first user query.
2. Delete the ones that ask the model about banned topics or about properties of the AI assistant.
3. Get LLMs to turn-wise generate user–assistant trajectories, instructed to answer naturally, but answer “****” if they cannot answer without introducing banned topics.
4. Apply another filter that removes unclean trajectories according to an LLM judge, and discard trajectories with “****” anywhere.
5. Add back in hand-written identity-anchoring data that tells the model basic aspects of its identity, without biasing it.
6. Train the pretrained model on these SFT trajectories.

Filtering Results

	LLM judges	Embedding model
Tier 1	1%	1%
Tier 2	5%	20%
Tier 3	50%	80%

False positive rate by tier, with thresholds configured to have a 0% false negative rate on held-out hand-labeled samples, with safety margin.

Tier 1 filters out philosophy of mind and explicit talk of consciousness. The idea being that this is filtering out the answers to all the direct questions we might want to ask.

Tier 3 filters out all documents where entities are having experiences. This is a pretty broad category, since very much text has experiential content. We erred on the side of caution in all stages of the process, and a lot of samples are hard to judge objectively, so we ended up with a pretty high false positive rate.

Tier 2 is a more surgical one in between 1 and 3, filtering out documents that *reify* experience. This means writing where entities are commenting on their own experience, or reflecting on it, making a distinction between the objects the experiences are about and the experience itself. The reasoning being that this is a fundamental mental move that underlies philosophical analyses of consciousness, without being as all-encompassing and mundane as experiential self-reports are in general.

The false positive rate is high for tier 3 because the category is somewhat vague, and we made conservative decisions in creating the validation data (classifying a lot of stuff as containing experience).

Train Results

So far we've trained three models:

1. annulus-1.4b-control
2. annulus-1.4b-reif
3. annulus-2.5b-reif

the first being unfiltered and trained on the same clean SFT data, the second two being trained on pretraining data filtered for tier 1 (PoM) and tier 2 (reification). These models were created mostly to test the filtering pipeline.

We can say pretty confidently that the philosophy of mind filtering worked — the filtered models simply don't know anything about philosophy of mind, even the 2.5b one. And the 1.4b control model does know a fair bit of stuff about PoM.

The reification filter is a bit less clear. With the 1.4b reification-filtered model, it was hard to tell, because it was too stupid to carry out the discussions required to probe its intuitions. With the 2.5b it's unclear — we don't know, as it hasn't finished training, but hopefully it should have by the time this poster is up, and you should be able to ask Will about it or chat with it here!

Filtering didn't reduce the performance of the models on downstream capabilities like overall knowledge or GSM8K.

Chat with the Model



Chat with annulus-2.5b-reif

Remaining Work

The three models here are created to test out the filtering process and make sure it removes all (or at least the vast vast majority of) confounding data. Right now the pipeline is in a state where it can scale up to training $4\times 7\text{--}8\text{b}$ models, which will be trained on unfiltered data, filtered for tier 1, filtered for 1+2, and filtered for 1+2+3. Then evaluations can proceed and we will get the final results of the experiment. However, to be completely sure the filtering works as intended, before we commit to the large final runs, we are auditing the small models and the filtered data for a little while before we proceed.

Costs

The pipeline produces 1bn filtered tokens at a cost somewhere between 1–20 USD per billion tokens. Running the embedding model over the data takes ~ 1 USD, and ensembling embedding models produces modest gains to accuracy. Generating labels with LLM judges cost around the same as running the embedding models for our experiments, but the fraction of total cost it comprises goes down as you filter a larger corpus. The most aggressive filtering (removing tiers 1, 2, 3, with conservatively set thresholds) produces around 1 filtered token per raw fineweb-edu token.

Look at the Data

Scan to browse the SFT seed prompts, the identity-anchoring trajectories, and examples of documents with judge ratings and reasoning.

